

WEEK-2

```
Last login: Mon Aug 17 14:48:21 on ttys001
amrutha@Amruthas-MacBook-Air ~ % mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 9.7.1 Homebrew

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE bookflow_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> 
```

Figure 1: Selecting the BookFlow database

```
Database changed
mysql> DESCRIBE Books;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id | int | NO | PRI | NULL | |
| title | varchar(100) | NO | | NULL | |
| isbn | varchar(20) | YES | UNI | NULL | |
| published_year | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.002 sec)

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title | isbn | published_year |
+-----+-----+-----+-----+
| 1 | Python Programming | ISBN001 | 2022 |
| 2 | Database Systems | ISBN002 | 2021 |
| 3 | Machine Learning | ISBN003 | 2024 |
| 4 | Java Programming | ISBN004 | 2025 |
+-----+-----+-----+-----+
4 rows in set (0.000 sec)
```

Figure 2: Verification of the Books table structure and existing book records

```
[mysql> DESCRIBE Members;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| member_id  | int       | NO   | PRI | NULL    |       |
| full_name  | varchar(100) | YES |     | NULL    |       |
| email      | varchar(100) | YES | UNI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.002 sec)

[mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email      |
+-----+-----+-----+
| 101      | Amrutha  | amrutha@gmail.com |
| 102      | Rahul    | rahul@gmail.com   |
| 103      | Priya    | priya@gmail.com   |
+-----+-----+-----+
3 rows in set (0.000 sec)
```

Figure 3: Verification of the Members table structure

```
[mysql> DESCRIBE Loans;
+-----+-----+-----+-----+-----+-----+
| Field      | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| loan_id    | int  | NO   | PRI | NULL    |       |
| member_id  | int  | YES  | MUL | NULL    |       |
| book_id    | int  | YES  | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.002 sec)

[mysql> SELECT * FROM Loans;
+-----+-----+-----+
| loan_id | member_id | book_id |
+-----+-----+-----+
| 1       | 101      | 1       |
| 2       | 102      | 2       |
| 3       | 103      | 3       |
+-----+-----+-----+
3 rows in set (0.003 sec)
```

Figure 4: Verification of the Loans table structure and loan records

```
mysql> SELECT
  ->     m.full_name AS Member_Name,
  ->     b.title AS Book_Title
  -> FROM Loans l
  -> INNER JOIN Members m
  ->     ON l.member_id = m.member_id
  -> INNER JOIN Books b
  ->     ON l.book_id = b.book_id;
+-----+-----+
| Member_Name | Book_Title |
+-----+-----+
| Amrutha     | Python Programming |
| Rahul       | Database Systems   |
| Priya       | Machine Learning   |
+-----+-----+
3 rows in set (0.001 sec)
```

Figure 5: INNER JOIN displaying member names and the books borrowed

```
mysql> SELECT
->     published_year,
->     COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
2021	1
2022	1
2024	1
2025	1

4 rows in set (0.003 sec)

Figure 6: Grouping books by publication year and counting the total number of books

```
[mysql> DESCRIBE Donation_History;
```

Field	Type	Null	Key	Default	Extra
donation_id	int	NO	PRI	NULL	
book_id	int	YES	MUL	NULL	
category	varchar(50)	YES		NULL	
donation_date	date	YES		NULL	

4 rows in set (0.002 sec)

```
[mysql> SELECT * FROM Donation_History;
```

donation_id	book_id	category	donation_date
1	4	Programming	2026-08-03

1 row in set (0.003 sec)

Figure 7: Verification of the Donation_History table structure and existing donation records

```
mysql> CREATE INDEX idx_books_isbn
```

```
[> ON Books(isbn);
```

Query OK, 0 rows affected, 1 warning (0.037 sec)

Records: 0 Duplicates: 0 Warnings: 1

Figure 8: Creating an index on the ISBN column for faster book searches

```
mysql> SELECT *
      -> FROM Books
      -> WHERE isbn = 'ISBN003';
```

book_id	title	isbn	published_year
3	Machine Learning	ISBN003	2024

```
1 row in set (0.002 sec)
```

Figure 9: Searching for a book using its indexed ISBN

```
[mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	Python Programming	ISBN001	2022
2	Database Systems	ISBN002	2021
3	Machine Learning	ISBN003	2024
4	Java Programming	ISBN004	2025

```
4 rows in set (0.035 sec)

mysql> SELECT * FROM Members;
```

member_id	full_name	email
101	Amrutha	amrutha@gmail.com
102	Rahul	rahul@gmail.com
103	Priya	priya@gmail.com

```
3 rows in set (0.001 sec)

mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id
1	101	1
2	102	2
3	103	3

```
3 rows in set (0.018 sec)

mysql> SELECT * FROM Donation_History;
```

donation_id	book_id	category	donation_date
1	4	Programming	2026-08-03

```
1 row in set (0.003 sec)
```

Figure 10: Final Books table ,Members table ,Loans table and Donation_History table